

ARUN KOTHANDARAMAN

DATA ANALYST | DATA ANALYTICS | BUSINESS INTELLIGENCE | DATA VISUALIZATION | REPORTING & INSIGHTS

Flagstaff, AZ | +1 (201) 490-2364 | a.kothandaraman1994@gmail.com | [LinkedIn](#) | [GitHub](#)

Summary

Data Analyst with **7+ years of experience** in data analysis, business intelligence, reporting, statistical analysis, and predictive analytics across **banking, manufacturing, gaming, and e-commerce**. Strong expertise in **SQL, Python, Power BI, Tableau, Excel, A/B testing, and statistical analysis**, with hands-on experience in data extraction, cleansing, transformation, validation, reconciliation, exploratory analysis, **ad hoc analysis**, dashboard development, and reporting automation. Experienced in analyzing large datasets, developing KPI dashboards, identifying trends and data-quality issues, and translating business requirements into actionable insights for cross-functional stakeholders.

Technical Skills

Programming & Data Analysis: SQL, Python, R, Pandas, NumPy, Scikit-learn, Matplotlib, Exploratory Data Analysis, Data Cleaning, Data Transformation

Databases & Data Warehousing: PostgreSQL, MySQL, Oracle, SQL Server, Snowflake, MongoDB, AWS Redshift, Databricks

Business Intelligence & Visualization: Power BI, Tableau, DAX, Excel, Power Query, Pivot Tables, VBA/Macros, KPI Dashboards, Data Visualization, Data Storytelling

Statistical & Business Analytics: Descriptive Statistics, A/B Testing, Hypothesis Testing, Regression Analysis, K-Means Clustering, Churn Analysis, Customer Segmentation, Time-Series Analysis, Predictive Analytics, Ad Hoc Analysis

Data Quality & Processes: Data Validation, Data Reconciliation, Data Quality Analysis, Source-to-Target Mapping, Business Rules, KPI Validation, UAT, Reporting Automation, Anomaly Detection

Data Engineering & Tools: AWS S3, AWS EC2, AWS Redshift, Microsoft Azure, ETL Pipelines, Data Modeling, Apache Spark, Jupyter Notebook, Git, Jira, RapidMiner

Experience

DATA ANALYST | Sunwest Bank | Flagstaff, AZ, USA Jan 2026 – Present

Project: Banking Customer, Transaction & Financial Analytics Platform

- Analyzed **1M+ banking records** to identify customer, transaction, loan, and financial trends.
- Developed advanced **SQL queries** using CTEs, window functions, joins, and aggregations for data extraction and reporting.
- Built **Power BI dashboards with DAX** to track transaction, customer, loan, revenue, and operational KPIs.
- Automated reporting workflows using **Python, Pandas, and Power Query**, reducing manual reporting effort.
- Performed **data validation, reconciliation, and UAT** to ensure accuracy across banking datasets and reports.
- Conducted **A/B testing and statistical analysis** to evaluate customer engagement initiatives and support business decisions.
- Delivered **ad hoc analysis** to address customer, transaction, payment, and financial business questions.
- Used **AWS S3 and Redshift** to access and analyze large banking datasets for analytics and reporting.
- Identified **transaction anomalies and risk indicators** to support fraud and risk-analysis initiatives.

DATA ANALYST | Accenture | Chennai, India Jun 2020 –Aug 2024

Project: Manufacturing Operations & Production Analytics Platform

- Analyzed **3M+ production, inventory, quality, machine, and operational records using SQL and Python** to identify performance trends and improvement opportunities.
- Developed advanced **SQL queries** using CTEs, window functions, joins, and aggregations to analyze production variances, bottlenecks, inventory, and KPIs.
- Built interactive **Power BI dashboards with DAX** to track production, machine utilization, quality, inventory, downtime, and department KPIs.
- Automated data preparation and reporting using **Python, Pandas, Power Query, and VBA**, recovering **12+ analyst hours per week**.
- Developed demand forecasting models using **Python, Scikit-learn, ARIMA, and Prophet**, reducing overstock and stockout incidents by **20%**.
- Conducted **ad hoc analysis and A/B testing** to evaluate process improvements and support data-driven decisions.
- Built bottleneck analysis that identified recurring production slowdowns, contributing to an **18% reduction in operational downtime**.
- Analyzed raw-material, quality, and machine data to reduce **excess inventory by 22%**, defect rates by **14%**, and unexpected machine failures by **30%**.
- Used **Microsoft Azure** to support data storage, processing, and analytics workflows, while performing data reconciliation and validation across reporting datasets.

DATA ANALYST | MOBILE PREMIER LEAGUE (MPL) GAMES | Bengaluru, India Feb 2019 – Feb 2020

Project: Gaming Customer, Revenue & Engagement Analytics Platform

- Analyzed Zero Loop Time across the end-to-end player journey, measuring stage-level latency from app launch and game selection through battle/coin-range selection, matchmaking, and battle start.
- Identified high-wait-time stages and analyzed their relationship with player drop-offs and battle abandonment, helping product and technical teams prioritize player-experience improvements.
- Analyzed player monetization behavior across entry fees, battle selections, rewards, and gameplay activity to identify patterns influencing revenue and player participation.
- Built player funnel analyses across game discovery, battle selection, matchmaking, and gameplay stages to identify conversion bottlenecks and behavioral drop-offs.
- Analyzed event-level gameplay data using **SQL and Python to understand player** behavior, gameplay patterns, feature adoption, and engagement trends.
- Evaluated **A/B test** performance for gameplay and product changes by comparing engagement, conversion, retention, and monetization metrics across test and control groups.
- Analyzed player activity, session frequency, gameplay behavior, and inactivity patterns to identify behavioral signals associated with churn and retention.

- Built dashboards and recurring reports tracking gaming KPIs including **DAU/MAU, retention, conversion, engagement, monetization, and funnel performance**.
- Analyzed **50M+ player economy and transaction records using Snowflake SQL and Python** to identify monetization trends and revenue leakage, contributing to a **12% revenue uplift** worth approximately **\$2M**.
- Developed a **Random Forest churn prediction model using Python and Scikit-learn** on **5M+ user records**, achieving **80% accuracy** and supporting retention initiatives that improved quarterly retention by **5%**.
- Analyzed **10M+ player sessions using SQL and Python** to identify engagement patterns and feature adoption opportunities, contributing to a **15% increase in feature adoption**.
- Designed and analyzed **A/B tests across 3 monetization features involving 2M+ users**, applying statistical testing that contributed to a **9% increase in ARPU**.
- Built **SQL-based fraud detection analysis across 50M+ transactions**, reducing **fraudulent coin transfers by 40%**.

ASSOCIATE DATA ANALYST | GSN GAMES| Bengaluru, India

Oct 2017 – Feb2019

Project: Player Engagement & Monetization Analytics Platform

- Built a near-real-time Tableau monetization dashboard, refreshed every 10 minutes, tracking hourly KPIs including transactions, payers, **ARPT, DAU**, and revenue, with same-day and previous-week comparisons to monitor changes in player spending
- Developed SQL-based player **cohort retention analysis, tracking D1, D3, D7, D14, and D45 retention rates** to measure return behavior, pinpoint key drop-off periods, and understand long-term engagement patterns.
- Developed payer segmentation and clustering analysis using player spending characteristics including lifetime value, total transactions, purchase frequency, average revenue per transaction, and offer purchases.
- Analyzed payer behavior across lifetime and recent activity windows **to identify purchasing patterns and support more relevant store-pack, offer, and price-point targeting across player segments**.
- Developed user-level payer reactivation logic using historical payment frequency to identify previously paying players who exceeded their expected purchase interval, enabling targeted in-game offers to drive payer conversion and monetization.
- Segmented **1M+ users into 6 high-LTV personas using R and SQL**, supporting targeted campaigns that increased **feature adoption by 14%** and **ARPU by 8%**.
- Conducted **A/B testing for 20+ campaigns** using statistical and hypothesis-testing techniques to evaluate campaign performance and provide data-driven recommendations.
- Built **SQL-based behavioral segmentation pipelines identifying 200K+ lapsed payers**, supporting reactivation campaigns that improved **purchase frequency by 18%**.
- Developed **Python ETL pipelines to process 500M+ daily game events**, reducing data latency from **24 hours to approximately 2 hours**.
- Analyzed **1M+ player progression paths using SQL and Python with Markov Chain analysis** to identify behavioral trends and support datadriven product decisions.

DATA ANALYST INTERN | MIRRAW| Mumbai, India

Jun 2017 – Aug 2017

Project: E-Commerce Customer & Sales Analytics Platform

- Queried **500K+ purchase transactions using SQL and Python** and developed an Apriori association-rule model to identify cross-sell opportunities, contributing to a **12% increase in average basket size**.
- Segmented **500K+ customers using SQL and Python** into LTV groups, supporting retention campaigns that increased repeat purchases by **10%**.
- Conducted **ad hoc SQL and Python analysis** of customer behavior, product performance, pricing, sales trends, and conversion metrics.
- Performed price-elasticity analysis using **Python and statistical techniques**, supporting pricing decisions and contributing to a **6% improvement in margins**.
- Analyzed search-to-purchase funnels using **SQL** to identify customer drop-off points, supporting improvements that increased conversion by **8%**.
- Built a Random Forest model using **Python and Scikit-learn** to predict delivery delays, contributing to a **20% reduction in SLA breaches**.
- Applied collaborative-filtering techniques using **Python** to improve product recommendation analysis, contributing to an additional **5% increase in basket size**.

Projects

Global Superstore Revenue Intelligence Dashboard SQL | PostgreSQL | Python | Power BI | DAX | AWS

- Developed an end-to-end revenue analytics solution using SQL and Python to clean, transform, and analyze sales data.
- Built interactive Power BI dashboards with DAX to track revenue, profit, customer, regional, and product performance.
- Identified sales trends and performance gaps to support data-driven business decisions.

Banking Analytics SQL | PostgreSQL | Python | Power BI| AWS

- Analyzed customer, account, transaction, loan, card, and regional banking datasets using SQL and Python.
- Performed data cleaning, transformation, and exploratory analysis to identify customer and transaction trends.
- Developed Power BI dashboards to monitor KPIs and support business and financial analysis.

Heart Disease Risk Prediction R | Logistic Regression | Machine Learning

- Cleaned and analyzed healthcare datasets using R and performed exploratory data analysis.
- Applied feature analysis and Logistic Regression to identify factors associated with heart disease risk.
- Evaluated model performance and presented analytical findings through visualizations.

Pizza Sales Analysis | Power BI | SQL | Excel

- Analyzed sales transactions using SQL and Excel to identify revenue, order, product, and customer trends.

- Created an interactive Power BI dashboard with KPIs, sales trends, product categories, and order analysis.
- Identified best-selling and underperforming products to support sales and inventory decisions.

Hotel Analysis| Power BI | Excel

- Analyzed hotel revenue, occupancy, bookings, cancellations, room categories, and customer data.
- Developed an interactive Power BI dashboard to track operational and revenue KPIs.
- Identified booking patterns and performance trends to support revenue and operational planning.

HR Analysis| Power BI | Excel

- Analyzed employee data including attrition, demographics, salaries, departments, and job roles.
- Built an interactive Power BI dashboard to monitor workforce KPIs and employee trends.
- Identified key attrition patterns to support workforce planning and retention analysis.

Hospital Emergency Room Dashboard| Power BI | Excel

- Analyzed emergency room data including patient volume, wait times, admissions, referrals, and satisfaction.
- Developed an interactive Power BI dashboard to monitor patient-flow and service-performance KPIs.
- Identified operational trends to support resource allocation and process improvement.

Education

- **M.S. in Business Analytics| Jan 2025 - May 2026** | Northern Arizona University – W. A. Franke College of Business| Flagstaff, AZ
- **Post Graduate Program in Data Science, Big Data & Business Intelligence |May 2016 - May 2017|** Aegis School of Business – IBM Certified| Mumbai, India
- **Bachelor of Engineering in Mechanical Engineering | May 2012 - Apr 2016|** Anna University – College of Engineering Guindy| Chennai, India